

Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Software Development Engineer, EC2 Network Provisioning** with **Amazon Web Services**. I bring over three years of professional experience building distributed, asynchronous backend systems at global scale, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across two engineering roles, I have built and delivered production-grade distributed systems that directly reflect the technical depth this position demands — designing, implementing, testing, and operating large-scale, multi-tiered, asynchronous services end-to-end. As a Software Engineer at Sperse, I designed a distributed, multi-tiered CRM orchestration platform in Python (LangGraph, FastAPI) that asynchronously routes requests across 40+ specialized sub-agents for 10,000+ users — with object-oriented service decomposition, tenant-aware access control, and OpenTelemetry monitoring instrumented end-to-end. Alongside it, I engineered asynchronous, fault-tolerant workflow execution with React Flow, TypeScript, and Temporal, enabling large-scale durable pipelines with multi-threaded task dispatch, live monitoring via Redis, and real-time reporting from a centralized dashboard. To keep these distributed systems production-ready, I built a FastMCP MCP server exposing 160+ backend actions to distributed AI agent fleets with multi-threaded dispatch, Pydantic schema validation, and OpenTelemetry tracing and alarming across multi-tenant deployments — and wired those systems into GitHub Actions CI/CD pipelines with trace-driven evaluation suites that gate every release through alarming thresholds before reaching production.

Prior to Sperse, as a Software Engineer at Qualitest, I built backbone infrastructure running at scale. I designed a large-scale, cloud-agnostic Storage Service in Java and Spring Boot — applying object-oriented design with a pluggable adapter pattern backed by PostgreSQL and Redis — handling distributed file operations across Google Cloud Storage, S3, and MinIO via expiring signed URLs, giving me hands-on experience with multi-tiered, multi-threaded Java service design of exactly the kind EC2 Network Provisioning engineers build daily. I also orchestrated asynchronous distributed jobs with AWS Step Functions and Lambda — implementing retry logic and fault recovery — eliminating timeout failures across 100K+ weekly executions with CloudWatch monitoring and alarming. This operational experience drove continuous improvement through rigorous observability: optimizing PostgreSQL with window functions, materialized views, and targeted indexes, and surfacing metrics through interactive dashboards for reporting and stakeholder visibility. These roles together have prepared me to own the full software development lifecycle — requirements gathering, system design, implementation, code reviews, testing, and operational excellence — and to contribute meaningfully to EC2 Network Provisioning's core engineering responsibilities.

Beyond my professional roles, my work on Dark Factory — a distributed coding-agent orchestrator built on dependency-aware DAGs with asynchronous multi-machine execution, fault-tolerant retries, and a multi-tiered scheduling layer for failure isolation — reflects the same system design mindset this team values: building reliable, self-healing distributed systems that operate without manual intervention, and relentlessly driving improvements to simplify and harden the infrastructure at their core.

I have always been passionate about **customer obsession** and **operational excellence** — two principles at the heart of Amazon's mission to be Earth's most customer-centric company — and envision myself contributing my talent to the furthering of Amazon's goal of building the world's most reliable, scalable cloud infrastructure. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan